

SASTRA DEEMED UNIVERSITY

(A University under section 3 of the UGC Act, 1956)

End Semester Examinations

May 2023

Course Code: MAT102R01

Course: ENGINEERING MATHEMATICS - II

QP No. :UA186-A

Duration: 3 hours

Max. Marks:100

PART - A

Answer all the questions

10 x 2 = 20 Marks

1. Find the radius of curvature of the plane mirror.
2. Examine the nature of stationary point (1,2) of the function $f(x, y)$ if, $f_{xx} = 6x, f_{xy} = 0, f_{yy} = 6y$.
3. Solve $p^3x - p^2y - 1 = 0$.
4. Show that the family of parabolas $y^2 = 4c(x + c)$ is self-orthogonal.
5. Prove that $\nabla(r^n) = nr^{n-2}\vec{r}$, where $\vec{r}$ is the position vector at any point.
6. Find the equation of tangent plane at (1,0,2) to the surface $x^2y + 2xz^2 = 8$.
7. The temperature at a point (x, y, z) is given by $T = x^2 + y^2 - z$. The mosquito located at (1,1,2) desires to fly in such a direction that it will warm as soon as possible. In what direction should it fly?

8. Prove that the area bounded by a simple closed curve C is given by $\frac{1}{2} \int_C (x dy - y dx)$.
9. Using Stoke's theorem, prove that $\text{div} (\text{curl } \vec{F}) = 0$, where $\vec{F}$ is a vector point function.
10. The work done by a conservative force when it moves a particle around a closed curve is _____.

PART - B

Answer all the questions

4 x 5 = 20 Marks

11. a) Show that the evolute of the cycloid $x = a(\theta - \sin \theta)$,
 $y = a(1 - \cos \theta)$ is another cycloid. (8)
- b) Find the extreme values of the function $\sin x + \sin y + \sin(x + y)$, $0 \leq x, y \leq \frac{\pi}{2}$. (7)

(OR)

12. a) Find the envelope of the system of lines $\frac{x}{l} + \frac{y}{m} = 1$, where l and m are connected by the relation $\frac{l}{a} + \frac{m}{b} = 1$ (l, m are parameters). (8)
- b) Find the curvature of the curve $r^n = a^n \sin n\theta$ in terms of r . (7)
13. a) Uranium disintegrates at a rate proportional to the amount then present at any instant. If M_1 and M_2 grams of uranium are present at times T_1 and T_2 respectively, show that half-life of the uranium is $\frac{(T_2 - T_1) \log 2}{\log M_1 - \log M_2}$. (8)
- b) Solve, by the method of variation of parameters,
 $\frac{d^2 y}{dx^2} - 2 \frac{dy}{dx} + y = e^x \log x$ (7)

(OR)

14. a) Solve, $(1+x)^2 \frac{d^2y}{dx^2} + (1+x) \frac{dy}{dx} + y = 2 \sin[\log(1+x)]$ (8)

b) A voltage Ee^{-at} is applied $t = 0$ to a circuit containing inductance L and resistance R is given by the differential equation $L \frac{di}{dt} + Ri = Ee^{-at}$. Show that the current at any time t is $\frac{E}{R-aL} (e^{-at} - e^{-\frac{Rt}{L}})$, assuming that $i = 0$ at $t = 0$. (7)

15. a) If r is the distance of the point (x, y, z) from the origin, prove that

$$\text{curl} \left(\vec{k} \times \text{grad} \frac{1}{r} \right) + \text{grad} \left(\vec{k} \cdot \text{grad} \frac{1}{r} \right) = 0. \quad (8)$$

b) Determine $f(r)$ so that the vector $f(r) \cdot \vec{r}$ is both solenoidal and irrotational. (7)

(OR)

16. a) Find the values of the constants a, b, c , so that

$\vec{F} = (axy + bz^3)\vec{i} + (3x^2 - cz)\vec{j} + (3xz^2 - y)\vec{k}$ may be irrotational. For these values of a, b, c , find also the scalar potential vector. (8)

b) Find the angle between $x^2 + y^2 + z^2 = 4$ and $z = x^2 + y^2 + 3$ at $(2, -1, 2)$. (7)

17. a) Use the divergence theorem to evaluate $\iint_S \vec{F} d\vec{s}$ where

$\vec{F} = \sin(\pi x)\vec{i} + zy^3\vec{j} + (z^2 + 4x)\vec{k}$ and S is the surface of the box with $-1 \leq x \leq 2, 0 \leq y \leq 1$ and $1 \leq z \leq 4$. (8)

b) Find the work done when a force

$\vec{F} = (x^2 - y^2 + x)\vec{i} - (2xy + y)\vec{j}$ displaces a particle in xy -plane from $(0,0)$ to $(1,1)$ along the curve (i) $y = x$, (ii) $y = x^2$. Comment the answer. (7)

(OR)

18. Verify Green's theorem in the xy plane for

$\int_C [(xy + y^2)dx + x^2dy]$, where C is the closed curve of the region? bounded by $y = x$ and $y = x^2$.

PART - C

Answer the following

1 x 20 = 20 Marks

19. Solve the differential equation $\frac{dy}{dx} = -\frac{5x+4y}{4x+5y}$, $y(0) = \sqrt{\frac{8}{5}}$. Use this solution obtained, determined the points on the curve which is at maximum and minimum distance from the origin.
